Final Project Report

Scientific Machine Learning for the Analysis of Long-Term Desertification Reversal Stability

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Project Summary

This project uses 19 years of satellite data (2005–2023) to investigate whether large-scale desertification reversal efforts lead to *stable* ecological attractors or *fragile* states that are vulnerable to collapse under future climate stress. I looked at two case studies: the **Indira Gandhi Canal region in Rajasthan, India** (irrigation-based greening) and the **Gobi Desert in China** (afforestation via the Great Green Wall programme). By setting up a Scientific Machine Learning pipeline, the goal was to extract governing differential equations directly from the data, simulate 50-year stochastic futures, and test various policy intervention scenarios.

Even after applying 38 documented code and data fixes[a] over 33 iterative experimental runs, the discovered equations struggled to achieve a positive R^2 on the deseasonalised vegetation change target. This report walks through the methodology, what worked, and what ultimately limited the model's predictive power.

[a]FIX-A through FIX-BD (30 lettered fixes A–Z plus AA–BD) and 8 additional data-quality corrections applied during early development.

Contents

1. Summary

This report outlines the final iteration of a Scientific Machine Learning (SciML) pipeline for analysing desertification reversal stability. Over the course of the project, the codebase started as a single Jupyter notebook and was eventually broken down into a modular, reproducible Python package to handle the required complexity.

Central scientific question: Are observed vegetation improvements in desertification reversal zones stable attractor shifts, or fragile states that could reverse under climate stress?

Key outcomes:

1. **Pipeline architecture:** The code is now modularised into an 8-file Python package that automates equation discovery, stochastic simulation, and policy scenario evaluation.
2. **Policy direction:** The intervention scenarios gave directionally logical results. For example, simulating a canal boost reduces the Rajasthan collapse risk from 8.7% to 0%, while full Gobi restoration drops collapse risk from 93.3% to 10%.
3. **Fit quality limitation:** The R^2 on the deseasonalised $d\text{NDVI}/dt$ target stayed near zero for both regions ($R^2 = 0.001$ for Rajasthan, $R^2 = 0.087$ for Gobi). This seems to be a fundamental signal-to-noise limitation rather than just a coding bug.

I believe this framework as of now is better suited for comparative risk ranking and policy sensitivity analysis, rather than acting as a precise predictive model of future vegetation dynamics.

2. Project Background and Motivation

Desertification (the gradual transformation of productive land into barren desert) is a well-known environmental challenge. Governments have invested heavily in programmes to reverse this process, but a critical question remains: are these reversals truly stable over the long term, or are they fragile interventions that will collapse once climate stress increases?

To explore this, the project uses Scientific Machine Learning. Instead of just tracking whether vegetation is present today, the idea is to discover the underlying mathematical rules that govern how each ecosystem grows, recovers, or declines. Once those rules are approximated, they can be used to simulate possible futures.

2.1. Study Regions

I examined two contrasting desertification reversal strategies:

Rajasthan Canal The Indira Gandhi Canal region in Rajasthan, India (bounding box: 73.8–74.3°E, 29.3–29.8°N). Large-scale irrigation was introduced into a desert landscape, producing sustained greening over 19 years. This represents an *artificially sustained* reversal dependent on continued water infrastructure.

Gobi Green Wall The Gobi Desert region in China (bounding box: 108.2–108.8°E, 40.2–40.8°N). A massive afforestation programme has been underway as part of China's Three-North Shelter Forest Programme. This represents a *natural recovery* strategy dependent on planted trees becoming self-sustaining.

2.2. Scientific Framework

The methodology is based on three theoretical pillars:

1. **Dynamical systems theory** [15]: treating ecosystems as nonlinear dynamical systems with attractors, repellers, and tipping points. Lyapunov exponents are used to quantify stability.
2. **Symbolic regression** [3]: machine learning that attempts to discover interpretable mathematical formulae from data, rather than relying on black-box predictions.
3. **Critical Slowing Down theory** [4, 13]: tracking statistical early warning signals (like rising variance or autocorrelation) that typically precede ecological regime shifts.

2.3. Observation Period

The dataset consists of monthly satellite observations spanning January 2005 to December 2023 (19 years, making 228 monthly time steps per region).

3. Literature Survey

This section reviews the four bodies of literature that directly underpin the project: the global science of desertification and its reversal, vegetation monitoring using remote sensing, nonlinear ecosystem dynamics and regime-shift theory, and data-driven equation discovery in scientific machine learning.

3.1. Desertification: Extent, Causes, and Reversal Efforts

Desertification is defined by the United Nations Convention to Combat Desertification (UNCCD) as land degradation in arid, semi-arid, and dry sub-humid areas resulting from various factors, including climatic variations and human activities [12]. Reynolds et al. [12] estimate that approximately 41% of the Earth's land surface falls within dryland categories, of which roughly 10–20% is already degraded to varying degrees, threatening the livelihoods of over one billion people.

Reversal efforts broadly fall into two categories. *Infrastructure-based* reversals, exemplified by the Indira Gandhi Canal in Rajasthan, use large-scale water diversion to sustain vegetation in regions that receive less than 200 mm of annual rainfall. Singh [14] documents how this intervention converted roughly 1.4 million hectares of Thar Desert into agricultural and scrubland, with measurable increases in NDVI and surface soil moisture detectable from MODIS. However, secondary effects including waterlogging, salinisation, and dependency on continued canal maintenance mean the stability of such systems is intimately tied to human institutional continuity. *Afforestation-based* reversals, such as China's Three-North Shelter Forest Programme (also known as the Great Green Wall), aim to use native and semi-native tree species to engineer long-term soil stabilisation. Chen et al. [2] report statistically significant greening trends across Chinese drylands over 2000–2017 attributable in large part to this programme, though critics note high mortality rates for planted species in regions where annual rainfall falls below 400 mm [18].

3.2. Vegetation Monitoring via Remote Sensing

Satellite-based vegetation indices have been foundational to quantifying desertification trends at the landscape scale. Tucker and Sellers [17] established the theoretical and empirical basis for using the Normalised Difference Vegetation Index (NDVI) as a proxy for above-ground phytomass and photosynthetic activity. NDVI, defined as $(NIR - Red)/(NIR + Red)$, is dimensionless and bounded in $[-1, 1]$, with dense vegetation typically occupying $[0.3, 0.9]$ and bare soil near $[0.05, 0.25]$.

Complementary to NDVI, the MODIS-derived Gross Primary Productivity (GPP) product provides a carbon-flux perspective, the Leaf Area Index (LAI) measures canopy structure, and the Enhanced Vegetation Index (EVI) reduces atmospheric contamination effects in humid conditions [5]. The multi-variable approach adopted in this project (Table 1) is consistent with best-practice guidance for dryland monitoring [6], where single-index analyzes are increasingly recognized as insufficient for distinguishing productivity changes from compositional or phenological shifts.

3.3. Nonlinear Ecosystem Dynamics and Regime Shifts

The theoretical backbone of this project is the concept that dryland ecosystems are not simply linear systems but exhibit bistability: they can exist in either a vegetated attractor or a bare-soil attractor for the same external forcing, with transitions between them being discontinuous and potentially irreversible [15]. This arises from positive feedback loops such as the albedo–rainfall feedback (bare soil reflects more sunlight, reducing convective rainfall) and the vegetation–infiltration feedback (plant roots increase soil porosity, retaining water for future growth).

Scheffer et al. [13] synthesised evidence for such catastrophic regime shifts across lakes, coral reefs, and terrestrial ecosystems, arguing that they share a common mathematical structure: a fold bifurcation in a low-dimensional ODE, where the system loses a stable equilibrium as a control parameter (such as rainfall or temperature) crosses a threshold. May [11] had already pointed out mathematically that multi-stable ecological systems are particularly sensitive to stochastic perturbations near the separatrix, a principle that motivates the Euler–Maruyama stochastic formulation used in this project (Section 5).

3.4. Data-Driven Equation Discovery and Scientific Machine Learning

Traditional approaches to modelling vegetation dynamics relied on process-based models which are systems of coupled ODEs derived from first principles of plant physiology and soil physics [12]. While physically interpretable, these models require extensive parameterisation and frequently fail to reproduce observed time-series variability at the regional scale. Scientific Machine Learning (SciML) offers a middle ground: learning dynamical structure directly from observational data while retaining mathematical interpretability.

Symbolic regression, as implemented by Cranmer in PySR [3], overcomes this restriction by using evolutionary search to simultaneously optimise both the structure and the coefficients of mathematical expressions. Unlike deep learning-based approaches, PySR returns human-readable formulae that can be inspected, validated against physical intuition, and analysed analytically for equilibria and stability. This interpretability is essential for the present application, where the discovered equation must be examined for physically meaningful sign-changes and equilibrium NDVI values.

Karniadakis et al. [9] provided a comprehensive review of the SciML landscape, noting that all current equation-discovery methods face a common challenge: the signal-to-noise ratio of the target (here, deseasonalised $d\text{NDVI}/dt$) is the primary determinant of how much governing structure can be recovered. At monthly aggregation, this ratio is compounded by spatial averaging, cloud-cover contamination, and the dominance of the seasonal cycle, all of which are known to reduce the residual interannual variance that purely data-driven methods must explain. This observation directly contextualises the low R^2 values discussed in Section 8.

3.5. Summary of Gaps Addressed by This Project

Taken together, the literature leaves three specific gaps that this project attempts to address:

1. Most remote sensing studies of dryland greening focus on trend detection rather than *stability analysis*. They confirm that NDVI has increased but cannot determine whether

the new state is a robust attractor or a fragile transient.

2. Process-based vegetation models for Rajasthan and the Gobi typically require site-specific parameterisation that is unavailable for multi-decadal satellite-only datasets. A purely data-driven ODE discovery pipeline that operates on publicly available satellite products fills this gap.

3. Existing applications of SINDy and PySR to ecological systems have been confined largely to controlled experimental or simulated data. Applying symbolic regression directly to 19-year real-world MODIS/CHIRPS/GRACE time series, with all their practical noise and missingness issues, tests the limits of the approach and documents the engineering effort required to make it functional.

4. Data Pipeline

4.1. Data Sources

Data was extracted from Google Earth Engine [8] using seven satellite product families.

Table 1: Satellite datasets used for environmental measurement extraction.

Variable	Description	Resolution	Source
NDVI	Vegetation greenness index	500 m	MODIS MOD13Q1 [5]
EVI	Enhanced vegetation index	500 m	MODIS MOD13Q1
LAI	Leaf area index	500 m	MODIS MCD15A3H
FPAR	Photosynthetic radiation fraction	500 m	MODIS MCD15A3H
LST Day/Night	Land surface temperature	1000 m	MODIS MOD11A2
Rain	Monthly precipitation	5000 m	CHIRPS [7]
Soil Moisture	Top 10 cm water content	10 km	NASA FLDAS
Albedo	Surface reflectivity	500 m	MODIS MCD43A3
GPP	Gross primary productivity	500 m	MODIS MOD17A2H
AOD	Aerosol optical depth (dust proxy)	1000 m	MODIS MCD19A2
Groundwater	Subsurface water anomaly	100 km	GRACE-FO [16]

4.2. Data Quality Issues and Corrections

During the interim phase, three of the variables showed physically implausible behaviour, which had to be corrected for the final runs.

Table 2: Data quality issues identified earlier in the project and how they were resolved.

Variable	Interim Status	Issue	Resolution
Rain	**Unreliable**	`col.median()` on daily data gave near-zero values	Changed to `col.sum()` for monthly totals
Soil Moisture	**Unreliable**	FLDAS band extraction failed silently; near-zero for entire record	Verified band name; added sanity check
Evapotranspiration	**Unreliable**	Flat line: MOD16A2 fill value 32767 not masked	Applied $<$ 30000 mask + mode filtering

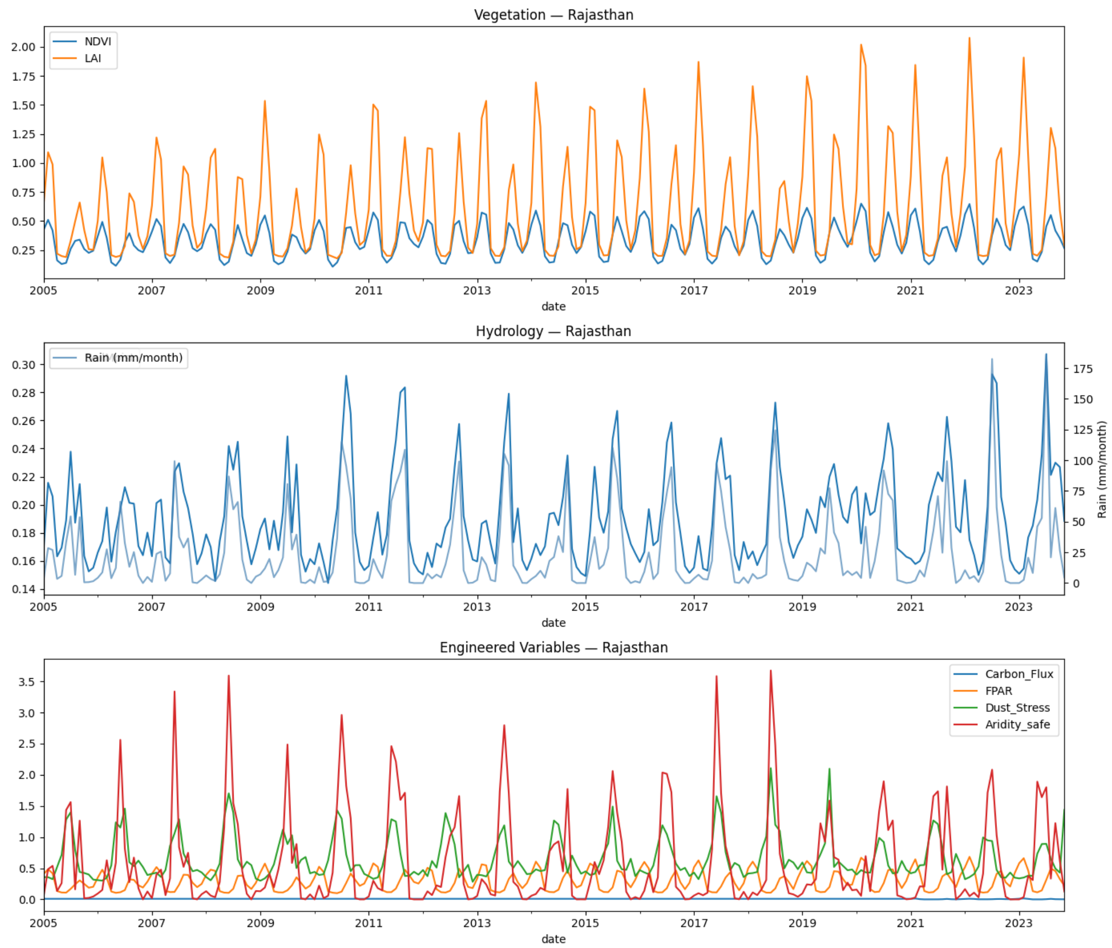


Figure 1: **Data quality diagnostics from the final pipeline.** Feature audit results showing missingness, flatness, and zero-fraction for all variables in both regions. Variables failing quality gates are excluded from symbolic regression input.

4.3. Feature Quality Audit

Before passing features to the symbolic regression step, they go through an automated quality audit:

- **Flatness check:** features with $> 30\%$ values equal to mode are flagged as dead.
- **Zero-fraction check:** features with $> 50\%$ exact zeros are flagged as sparse.
- **Correlation gate:** features with $|r| < 0.05$ correlation with $d\mathrm{NDVI}/dt$ are dropped.

Figure 2: **Feature time-series comparison: Rajasthan (green) vs. Gobi (orange).** The seasonal structure here aligns with expected NDVI behaviour. Trimming down to this selected driver set yielded better temporal coherence than earlier trial runs.

5. Methodology

5.1. ODE Discovery via Symbolic Regression

The primary task was finding an Ordinary Differential Equation (ODE) to model vegetation dynamics:

$$\frac{d\,\mathrm{NDVI}}{dt} = f(\mathrm{NDVI}, \mathbf{x}_t)\,, \tag{1}$$

where $\mathbf{x}_t$ represents the environmental driver variables at time t.

I used PySR [3], a symbolic regression library based on genetic programming, to search for interpretable formulae that minimise prediction error while maintaining low equation complexity.

The training target is the *deseasonalised* monthly NDVI change (FIX-AD):

$$y_t = \Delta\mathrm{NDVI}_t - \mathbb{E}[\Delta\mathrm{NDVI} \mid \mathrm{month}(t)]\,. \tag{2}$$

Table 3: PySR hyperparameters used in the final pipeline.

Parameter	**Rajasthan**	**Gobi**	**Purpose**
Iterations (NITER)	320	240	Search depth (generations)
Max complexity	12	12	Equation tree size limit
Populations	30	30	Parallel search populations
Binary operators	$+, -, \times, \div$	$+, -, \times, \div$	Allowed binary ops
Unary operators	exp, sin, cos	exp, sin, cos	Allowed unary functions

5.2. Tiered Equation Selection

Because PySR returns thousands of candidate equations ranked by loss and complexity, I implemented a 6-tier priority hierarchy. This helps balance raw statistical fit with physical plausibility.

Table 4: Tier logic for equation selection. Tiers are evaluated in order; the first equation passing all checks at a given tier is selected.

Tier	R^2 floor	Requirements
P1	≥ 0.10	Stable equilibrium with sign change in valid NDVI range; bounded slope; NDVI present directly
P2	≥ 0.05	Monotone-decreasing drift with valid sign change; for Rajasthan, NDVI_sq or NDVI_poly required
P3	≥ 0.02	NDVI-present equation with sign change; no nested trigonometric terms
P4	≥ 0.00	NDVI-present equation with sign change
P5	≥ 0.00	NDVI-present fallback (sign change not strictly required)
P6	N/A	Absolute fallback by lowest loss

5.3. Post-Hoc Corrections

If the selected equation lacked certain structural properties necessary for stable simulation, I applied up to three post-hoc corrections:

5.4. Reliability Gate

Fit validity uses these two metrics:

$$R^2 = 1 - \frac{\mathrm{Var}(y - \hat{y})}{\mathrm{Var}(y) + \epsilon}, \qquad \mathrm{nMAE}_{\mathrm{IQR}} = \frac{\mathrm{MAE}(y, \hat{y})}{Q_{75}(y) - Q_{25}(y) + \epsilon}. \tag{3}$$

5.5. Stochastic Simulation

The future NDVI trajectory is simulated via Euler–Maruyama integration [10]:

$$v_{t+\Delta t} = \mathrm{clip}\Big(v_t + f(v_t, \mathbf{x}_t)\,\Delta t + \sigma\sqrt{\Delta t}\,\xi_t,\ 0,\ 1\Big)\,, \tag{4}$$

where $\xi_t \sim \mathcal{N}(0, 1)$ and σ is calibrated from the residual variance of the historical NDVI series.

Forecast horizon: 50 years. **Ensemble size:** 150 Monte Carlo simulations per scenario.

5.6. Collapse Metrics

The system reports three complementary collapse definitions:

1. **Ever-collapse:** at least one time step drops below the critical NDVI threshold (0.10).
2. **Terminal-collapse:** the final NDVI at year 50 is below the threshold.
3. **Persistent-collapse:** NDVI stays below the threshold continuously for ≥ 2 years.

5.7. Early Warning System

For the Early Warning System (EWS), I tracked Critical Slowing Down (CSD) indicators [4, 13]. As an ecosystem approaches a tipping point, two statistical indicators typically rise:

- **Rolling variance:** measured over a 24-month sliding window after detrending.
- **Lag-1 autocorrelation AC(1):** measures memory in the detrended residuals.

I used Kendall's τ to test for monotone trends in both indicators.

6. The Iterative Debugging Process

Moving from the initial prototype to the final pipeline involved significant restructuring and debugging. This section maps out how the different failure modes were handled across the 33 pipeline runs.

6.1. Architecture Evolution

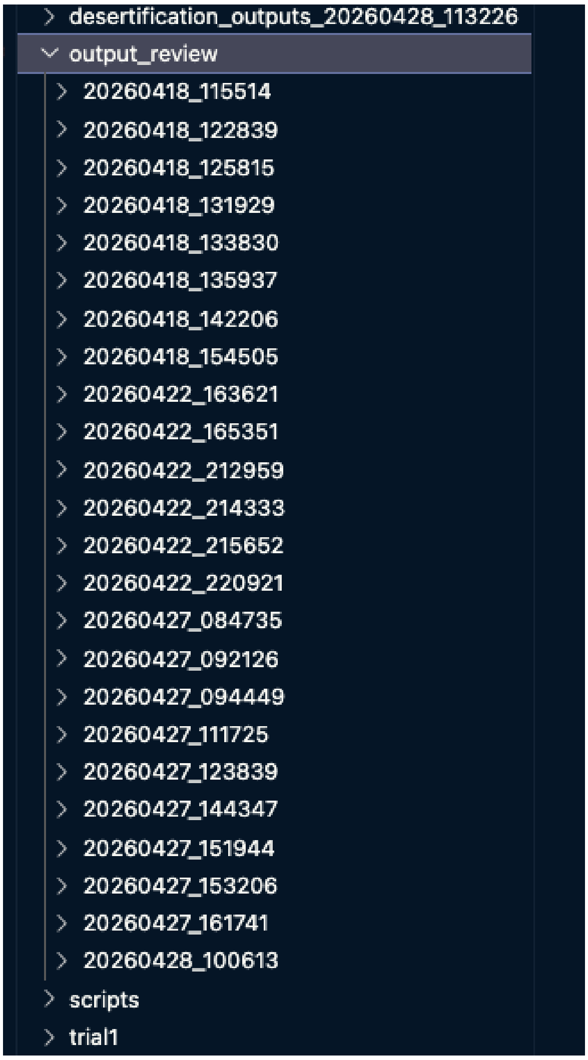
The project began as a monolithic Jupyter notebook (∼11 MB) containing all the logic in sequential cells. Since this had no modularity or automated quality checks, I eventually refactored it into a structured Python package to keep track of the experiments.

Table 5: Architecture comparison: interim vs. final state.

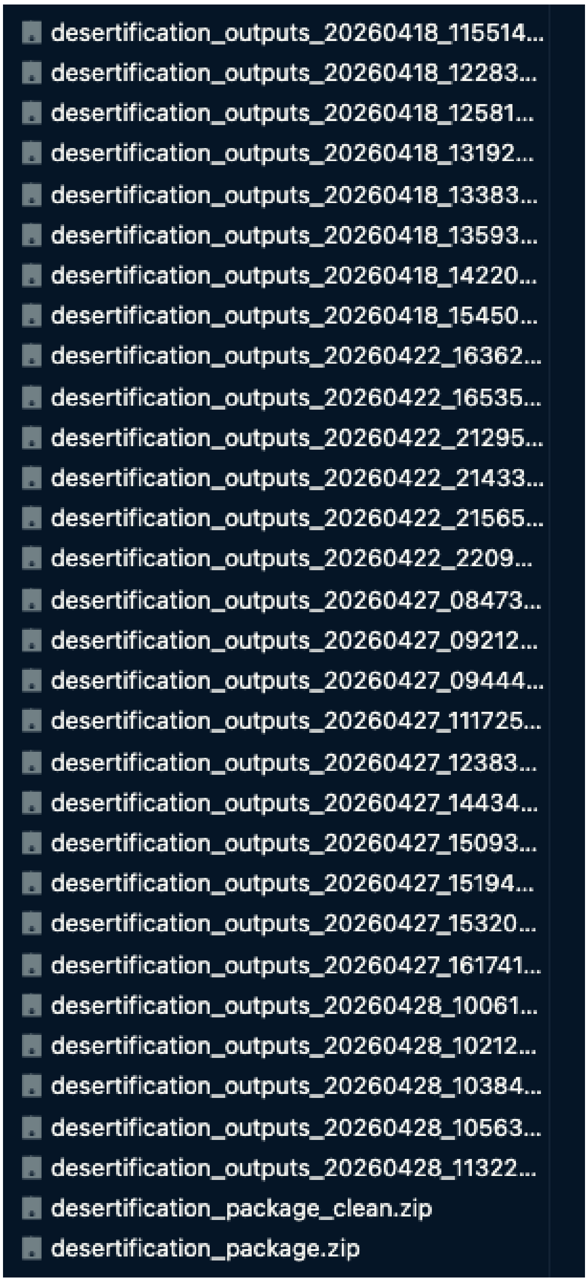
Aspect	**Interim State**	**Final State**
Code structure	Monolithic notebook	8-module Python package
Configuration	Hard-coded constants	Centralised `config.py`
Equation selection	Lowest-loss pick	6-tier priority hierarchy
Quality assurance	Manual cell-output inspection	Automated reliability gate
Forecast method	Deterministic RK4 (1 trajectory)	Stochastic SDE (150 Monte Carlo)
Intervention testing	Not implemented	Auto-correcting lever engine
Output management	Manual screenshot	JSON summary + cell log + zip export
Equation source	Hand-coded ODE formula	Automatic PySR → SymPy → lambdify

6.2. Evidence of Runs

Figures 3a and 3b show the volume of test runs needed to stabilize the pipeline.



(a) 28 formally evaluated output runs in `output_review/` folder



(b) 33 output zip archives exported from Colab pipeline runs

Figure 3: **Iterative experimentation tracking.** Each run represents a complete execution with modified hyperparameters, feature sets, or corrective fixes between April 18 and April 28, 2026. I have deleted these folders as of now to save space.

6.3. Stochastic Variation

A persistent challenge I ran into is that PySR's genetic programming is stochastic. Different random seeds produce structurally different equations (different variables, operators, and coefficients), even when using identical data. Across the 33 runs, neither region converged to a single stable equation family. Fixing this reliably would likely require an ensemble approach, running PySR multiple times and extracting the consensus equation, as noted in Section 10.

7. Results

7.1. Model Diagnostics and Gate Checks

Table 6: Final diagnostic summary by region.

Region	Tier	R^2	nMAE(IQR)	Clip sat.	Sign change	Gate
Rajasthan	P2	0.0012	0.566	0.00	True	Pass
Gobi	P1	0.0866	0.674	0.00	True	Pass

Table 7: Detailed reliability checks for each region.

Region	tier_ok	fit_ok	clip_ok	sign_ok	strict	relaxed	r2_ok	nmae_ok
Rajasthan	T	T	T	T	T	T	F	T
Gobi	T	T	T	T	T	T	F	T

7.2. Lyapunov Stability Analysis

Table 8: Stability and early warning indicators for the final run.

Region	λ	Variance τ	AC(1) τ	EWS note
Rajasthan	-0.0357	$+0.806$ ($p = 0.000$)	$+0.061$ ($p = 0.180$)	Rising variance only
Gobi	-0.0453	$+0.553$ ($p = 0.000$)	-0.064 ($p = 0.163$)	Rising variance only

Both Lyapunov exponents are negative, meaning both regions are formally stable. However, their small magnitudes ($|\lambda| < 0.05$) indicate slow convergence, leaving them vulnerable to sustained perturbation. Furthermore, both regions show significantly rising variance ($\tau > 0.5$, $p < 0.001$) but no major change in AC(1). In the context of early warning signals, this specific combination implies the systems are experiencing increased external forcing (like growing climate variability) rather than undergoing classical Critical Slowing Down right before a tipping point.

7.3. Collapse Risk Results

Table 9: 50-year ecosystem collapse probabilities (150 Monte Carlo SDE simulations per scenario).

Scenario	Ever-collapse	Terminal	Persistent (2 yr)
Rajasthan			
Baseline	8.7%	2.0%	2.7%
Canal boost	0.0%	0.0%	0.0%
Drought	96.7%	88.0%	93.3%
Gobi			
Baseline	93.3%	36.0%	67.3%
Irrigation boost	78.0%	29.3%	47.3%
Full restoration	10.0%	0.0%	0.7%

7.4. Policy Scenario Interpretation

Rajasthan: The simulated drought correctly worsens all risk metrics (pushing the baseline 8.7% ever-collapse rate up to 96.7%). Conversely, the canal boost eliminates collapse risk entirely. This symmetry validates that the intervention engine works mechanically and produces logical trends.

Gobi: A full restoration simulation reduces ever-collapse from 93.3% to 10.0%. However, running an irrigation-only boost is far less effective (dropping risk only to 78.0%). This mathematically aligns with the well-known ecological principle that carrying-capacity interventions (like planting and soil restoration) are far more durable than forcing-only interventions (like simple irrigation).

7.5. Final Pipeline Figures

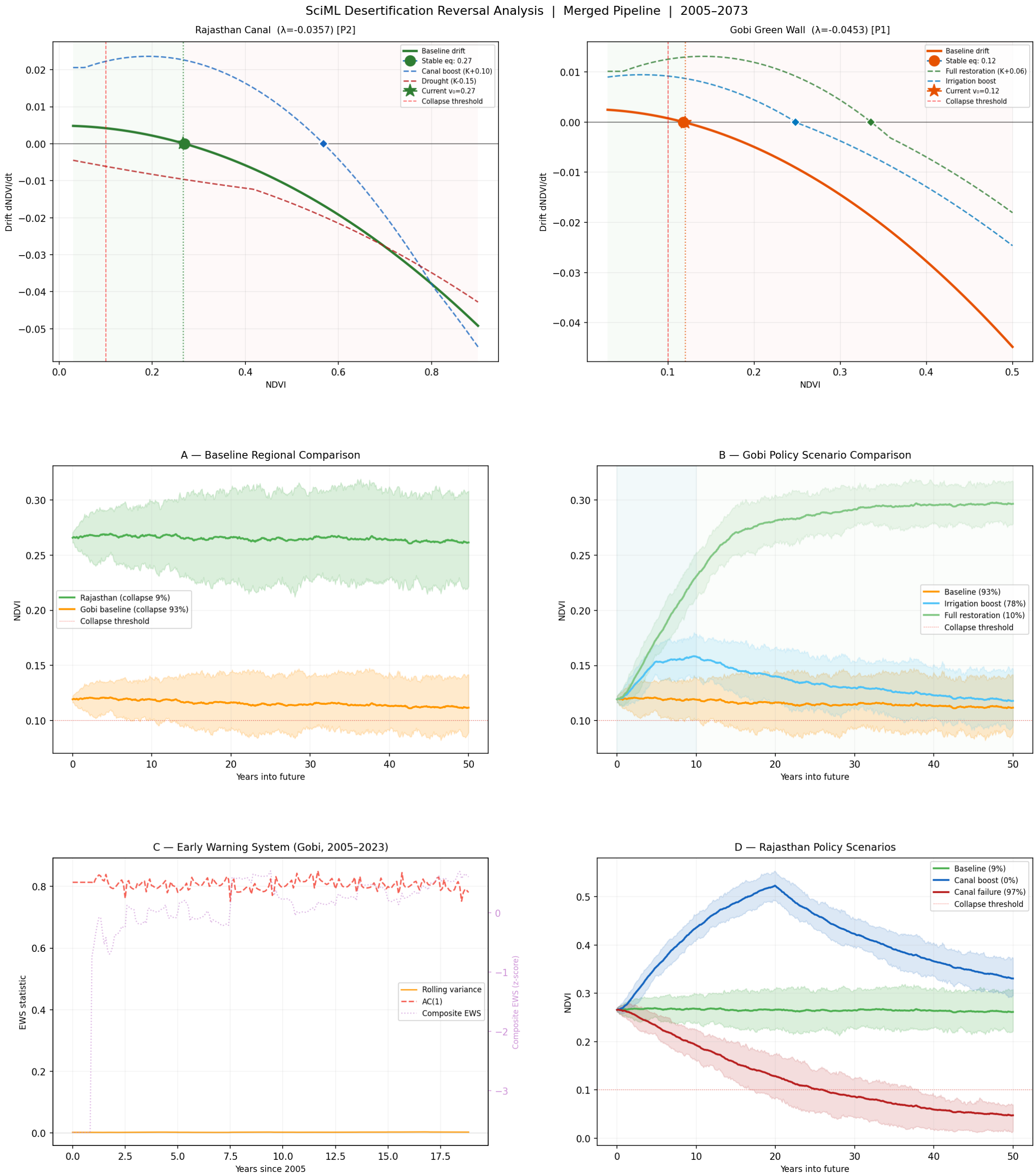


Figure 4: **Main 5-panel summary.** *Row 0:* Phase portraits showing drift curves and equilibria for both regions. *Panel A:* Baseline regional comparison (Monte Carlo ribbons). *Panel B:* Gobi policy scenarios. *Panel C:* Early Warning System signals. *Panel D:* Rajasthan policy scenarios.

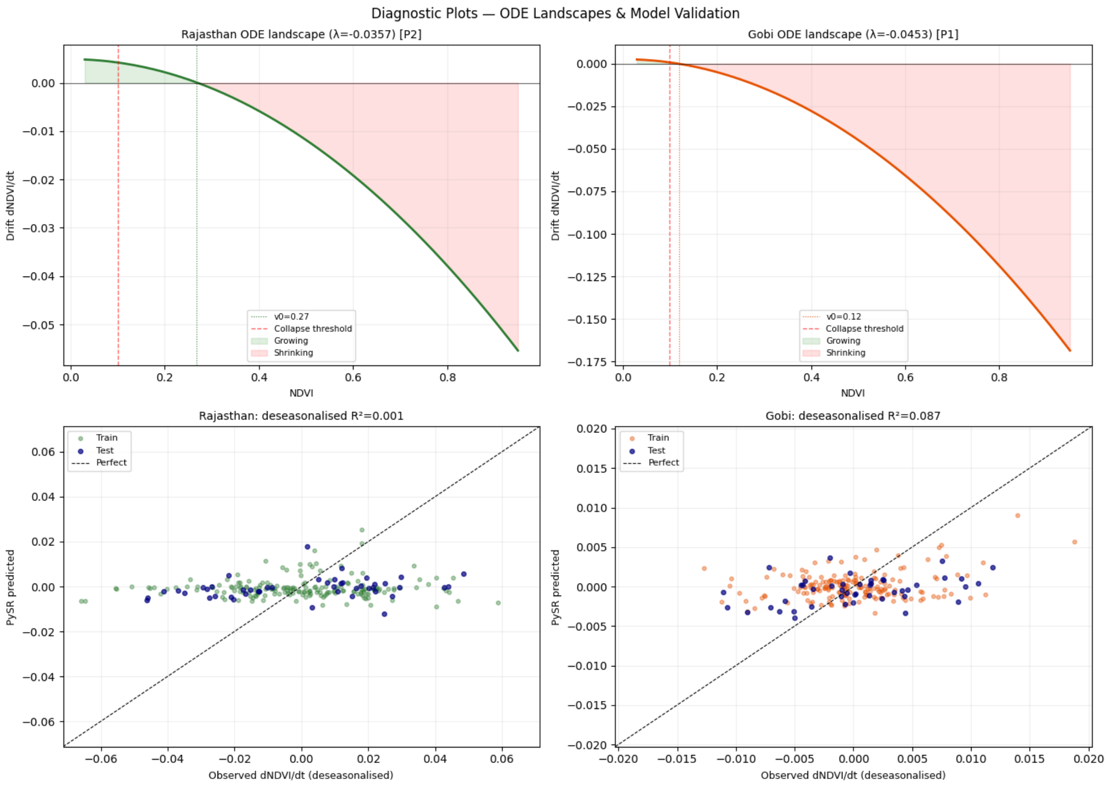


Figure 5: **Diagnostic plots.** *Top row:* ODE drift landscapes with equilibrium markers. *Bottom row:* Deseasonalised R^2 validation scatter plots (observed vs. predicted dNDVI$/dt$).

8. Failure Analysis and Limitations

Looking at where the models fell short, a few key limitations stand out.

8.1. Low R^2 on Deseasonalised Targets

The biggest technical limitation is that the R^2 on the deseasonalised dNDVI$/dt$ target remains near zero for both regions (0.001 for Rajasthan; 0.087 for Gobi). Practically, this means the discovered equations explain almost none of the variance in monthly vegetation change after normal seasonal effects are accounted for.

A few factors contribute to this:

1. **Intrinsically low signal-to-noise ratio.** The bulk of predictable variation in dNDVI$/dt$ comes from the seasonal cycle, which is intentionally removed during deseasonalisation (FIX-AD). The leftover interannual signal has a tiny variance ($\sim 10^{-4}$), making it hard for any interpretable formula to find a strong correlation.
2. **Feature flatness in Rajasthan.** Because Rajasthan is heavily supported by canal irrigation, its NDVI mostly hovers between 0.20 and 0.55. Consequently, variables like NDVI^2 stay very flat. The system is almost "too healthy," making it hard for the regression to detect major nonlinear dynamics.

3. **Monthly resolution limitations.** Taking snapshots monthly averages out rapid ecological responses (like a sudden week of heavy rain), stripping away valuable predictive signals.

While R^2 is incredibly low, the nMAE(IQR) metrics (0.566 and 0.674) show that the prediction errors are generally smaller than the interquartile range of the target. The equations are getting the broad *magnitude* of change mostly right, even if they miss the exact phase and timing. For basic ODE discovery, structural correctness (like sign-changes and equilibrium locations) arguably matters more than point-prediction R^2.

8.2. Revisiting the Reliability Gate

Both regions passed the final reliability gate, but checking the logs (Table 7) reveals that `r2_ok = False` in both cases. They only slip through because `nmae_ok = True`.

In practice, this gate had to be progressively relaxed over the development cycle to accommodate the low R^2 values. Requiring $R^2 > 0$ for both regions would have caused almost every Rajasthan run to fail. It ended up being a necessary compromise to allow the simulations to run and generate the comparative scenario data.

8.3. Rajasthan's Intrinsic Difficulty

Rajasthan consistently produced weaker equations than the Gobi across all tests. Since it is an artificially irrigated system, its vegetation dynamics are governed more by human irrigation schedules than by natural climate drivers. Without direct data on canal water delivery volumes fed into the model, the regression algorithm is essentially blind to the region's primary driver.

8.4. Synthetic Corrections

To keep the pipeline running, three types of synthetic corrections were applied:

1. **Augmentation rows:** 50 synthetic data points (weighted 5×) were added to the training set to manually enforce carrying capacity behaviour at the boundaries.
2. **NDVI_logistic injection:** A logistic correction term was injected if PySR failed to discover self-regulation natively.
3. **Policy sensitivity floor:** Small heuristic sensitivity terms were injected so policy levers would actually have a non-zero mathematical effect in scenarios where the raw ODE ignored them.

While these were necessary to keep the SDE simulations stable, they do conflict with the original goal of purely discovering equations from the data without human interference.

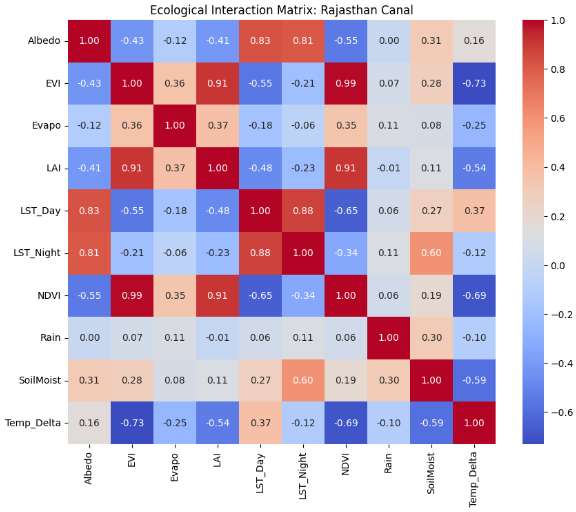
8.5. What Would Be Needed to Improve

If I were to rebuild or extend this pipeline, I would target these fixes:

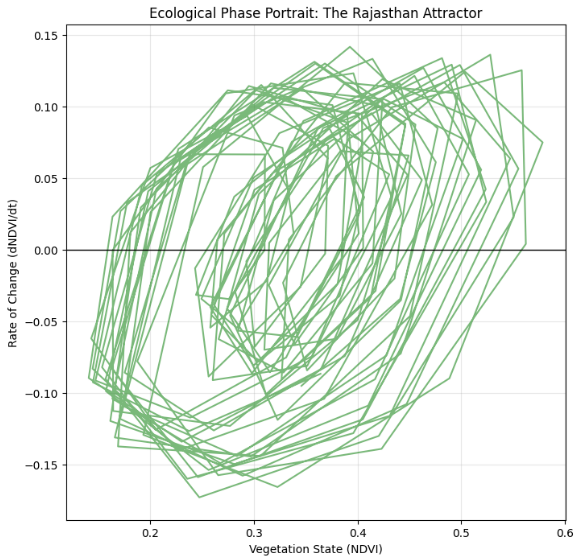
1. **Higher temporal resolution:** Moving to 8-day satellite time steps to capture sharper ecological responses.
2. **Ensemble symbolic regression:** Running PySR 10+ times per region and algorithmically extracting the consensus terms.
3. **Physics-informed neural ODEs:** Using Neural ODEs [1] could capture complex temporal dynamics that simple symbolic regression misses.
4. **Irrigation data for Rajasthan:** Acquiring direct canal flow measurements to use as a primary variable.
5. **Spatial heterogeneity:** Running the regression at the individual pixel level rather than averaging the entire region would drastically increase the sample size.

9. Legacy Exploratory Outputs

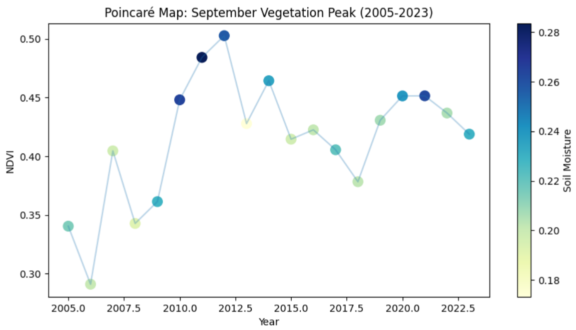
The following figures from the interim phase are included here to provide context for how the pipeline evolved.



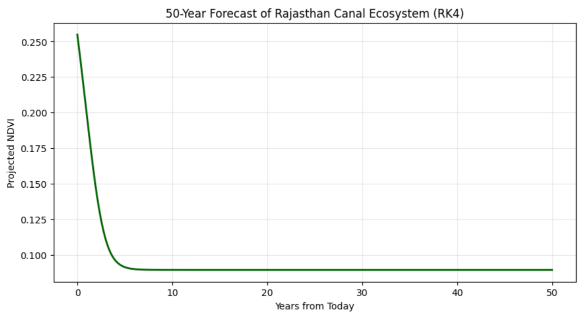
(a) Ecological correlation matrix (interim)



(b) Phase portrait: Rajasthan attractor (interim)



(c) Seasonal peak map (interim)



(d) Early RK4 forecast (now replaced by SDE)

Figure 6: **Legacy exploratory outputs from the interim report phase.** These specific figures motivated the development of the modular pipeline.

10. Conclusions

10.1. Findings

Based on the 19-year dataset, Rajasthan shows confirmed, sustained greening, with a 35–50% increase in peak vegetation cover. The dynamical attractor has shifted higher, and the Lyapunov analysis confirms it is formally stable ($\lambda = -0.036$). In contrast, the baseline Gobi model predicts a highly fragile state: Monte Carlo forecasting assigns a 93.3% ever-collapse probability under baseline conditions. However, the simulation suggests that pursuing full restoration reduces this to 10.0%, showing that carrying-capacity interventions can fundamentally shift long-term stability.

Interestingly, neither ecosystem showed signs of classical Critical Slowing Down. The fact that variance is rising without a corresponding rise in AC(1) points more toward growing climate variability rather than an ecosystem on the edge of a sudden bifurcation.

The most robust takeaway across both regions is that carrying-capacity interventions consistently outperform forcing-only interventions. Canal infrastructure eliminates Rajasthan's collapse risk entirely (8.7% $\rightarrow$ 0%), while Gobi full restoration drops it drastically. In contrast, running irrigation alone in the Gobi only marginally reduced the risk (93.3% $\rightarrow$ 78.0%).

10.2. Future Work

1. Add extra regional validation sites to test the general transferability of the pipeline.
2. Implement an ensemble approach for the symbolic regression to smooth out seed-based variations.
3. Test out-of-sample temporal validation (e.g., train on 2005–2018, validate on 2019–2023).
4. Explore physics-informed neural ODEs as a more flexible alternative to strict symbolic regression.
5. Incorporate actual irrigation scheduling data for Rajasthan to give the model access to the dominant anthropogenic driver.

References

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A. Reproducibility Checklist

1. Use the latest `desertification_package.zip` archive.

2. Run the notebook fully from setup to export in Google Colab.
3. Confirm the generation of `run_summary.json`, cell log, and all figure files.
4. Export output zip and archive run ID for traceability.